

QUESTION:12QUEUE USINGLINKEDLIST

main.c

Share

Run

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  struct Node {
4      int data;
5      struct Node *next;
6  };
7  struct Node *front = NULL;
8  struct Node *rear = NULL;
9  void enqueue(int value) {
10     struct Node *newNode;
11     newNode = (struct Node*)malloc(sizeof(struct Node));
12     if (newNode == NULL) {
13         printf("Memory Allocation Failed\n");
14         return;
15     }
16     newNode->data = value;
17     newNode->next = NULL;
18     if (front == NULL && rear == NULL) {
19         front = rear = newNode;
20     } else {
21         rear->next = newNode;
22         rear = newNode;
23     }
24     printf("%d inserted into queue\n", value);
25 }
26 void dequeue() {
```

Output

```
--- QUEUE USING LINKED LIST ---
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter value: 8
8 inserted into queue

--- QUEUE USING LINKED LIST ---
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 3
Queue elements are:
8

--- QUEUE USING LINKED LIST ---
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: |
```

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Run

```
27     struct Node *temp;
28     if (front == NULL) {
29         printf("Queue Underflow\n");
30         return;
31     }
32     temp = front;
33     printf("Deleted element: %d\n", front->data);
34     front = front->next;
35     if (front == NULL) {
36         rear = NULL;
37     }
38     free(temp);
39 }
40 void display() {
41     struct Node *temp;
42     if (front == NULL) {
43         printf("Queue is Empty\n");
44         return;
45     }
46     temp = front;
47     printf("Queue elements are:\n");
48     while (temp != NULL) {
49         printf("%d ", temp->data);
50         temp = temp->next;
51     }
52     printf("\n");
```

```
53 }
54 int main() {
55     int choice, value;
56     while (1) {
57         printf("\n--- QUEUE USING LINKED LIST ---\n");
58         printf("1. Enqueue\n");
59         printf("2. Dequeue\n");
60         printf("3. Display\n");
61         printf("4. Exit\n");
62         printf("Enter your choice: ");
63         scanf("%d", &choice);
64         switch (choice) {
65             case 1:
66                 printf("Enter value: ");
67                 scanf("%d", &value);
68                 enqueue(value);
69                 break;
70             case 2:
71                 dequeue();
72                 break;
73             case 3:
74                 display();
75                 break;
76             case 4:
77                 exit(0);
```

```
78             default:
79                 printf("Invalid Choice\n");
80         }
81     }
82     return 0;
83 }
```